

Problem Statement Analysis

Problem Statement #64 — Sustainability & Green Tech

Project: Disaster Relief Supply & Volunteer Coordinator

This document analyzes the professor-provided problem statement and converts it into a clear software-engineering understanding that can be used to prepare the required requirements table, UML use-case diagram, and use-case flow specification.

1. What the Problem Is About

The proposed system is an emergency disaster-management platform. Its purpose is to coordinate three closely connected relief activities: identifying critical supply needs at shelters, coordinating volunteers according to their skills, and tracking relief-vehicle deliveries.

The problem is therefore not simply about storing information. The system must help a Disaster Manager decide what needs attention, connect available resources with shelter needs, involve suitable volunteers, and maintain visibility of deliveries.

2. Main Objectives of the System

- **Manage shelter needs:** capture and organize requests for essential supplies such as water, blankets, and medical kits.
- **Match needs with resources:** compare incoming shelter requisitions with available central-warehouse supplies.
- **Prioritize relief:** ensure critical needs receive higher dispatch priority.
- **Coordinate volunteers:** use volunteer skill sets when assigning relief tasks.
- **Track deliveries:** maintain the status of relief vehicles and their deliveries.

3. Actors

Actor	Role in the System
Disaster Manager	Primary coordinator who manages shelter requests, checks resources, prioritizes deliveries, assigns volunteers, and monitors relief operations.
Relief Volunteer	Participant in relief operations who receives assigned work and provides availability/task-status information.

4. Core System Functions Identified

Area	System Function / Use Case	Purpose
Supply Management	Manage Shelter Supply Requests	Record and maintain requests for water, blankets, medical kits, and other relief supplies.
Inventory	Check Supply Availability	Determine whether requested quantities are available in the central warehouse.
Prioritization	Prioritize Relief Delivery	Give critical and urgent requests higher dispatch priority.
Volunteer Coordination	Assign Volunteers	Match volunteer skills and availability to relief tasks.
Delivery Tracking	Track Relief Vehicle Delivery	Monitor relief-vehicle delivery progress and status.
Volunteer Interaction	View Assigned Task / Update Availability / Update Task Status	Allow volunteers to see and maintain information related to their assigned work.

5. Functional Requirements Derived from the Problem

- **FR-001 — Manage shelter supply requests:** The system should allow shelter requirements for essential supplies to be recorded and managed.
- **FR-002 — Check warehouse availability:** The system should compare requests with available central-warehouse stock.
- **FR-003 — Prioritize deliveries:** The system should allocate delivery priority according to urgency and criticality.
- **FR-004 — Assign volunteers:** The system should use volunteer skill sets and availability when assigning relief work.
- **FR-005 — Track vehicle deliveries:** The system should track relief-vehicle delivery status.

6. Non-Functional Requirements Derived from the Given Guideline

- **NFR-001 — Performance/Security:** The dashboard should consume low network bandwidth and protect disaster-management information.
- **NFR-002 — Availability/Offline Support:** Essential GIS/disaster-mapping data should remain accessible through offline caching during network outages.

These two NFR directions are directly supported by the assignment's supplied NFR guideline, which specifically mentions low-bandwidth operation and offline GIS data caching, along with performance/security validation under simulated peak load.

7. Important UML Relationships

Relationship	Use Cases	Meaning
«include»	Prioritize Relief Delivery → Check Supply Availability	Checking supply availability is a required part of prioritizing a delivery.
«extend»	Report Emergency Shortage → Manage Shelter Supply Requests	Reporting an emergency shortage represents optional/conditional behavior associated with managing a request.

8. Recommended Core Use Case for the Flow Specification

Prioritize Relief Delivery is the strongest core use case for the one-page flow specification because it directly reflects the sample functional requirement supplied in the assignment: incoming shelter requisitions are matched against available warehouse supplies and a delivery priority is allocated.

- **Preconditions:** authorized Disaster Manager, registered shelter request, request details available, and current warehouse information available.
- **Main success:** select request → view requirements → check availability → evaluate urgency → assign priority → record priority → display result.
- **Alternate flow:** insufficient stock is detected → request is marked as a shortage → manager is alerted → alternative supply/replenishment can be arranged.
- **Postconditions:** priority is recorded and the request is ready for further relief-delivery processing.

9. Assignment Deliverables — What Must Be Submitted

Deliverable	Required Content
1. Requirements Table	Exactly 5 Functional Requirements (FR-001 to FR-005) and 2 Non-Functional Requirements (NFR-001 and NFR-002), each with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
2. UML Use-Case Diagram	All actors and primary use cases, with at least one «include» and one «extend» relationship.

Deliverable	Required Content
3. Use-Case Flow Specification	One core use case with Preconditions, Postconditions, Main Success Scenario, and one Alternate Flow.

10. Final Understanding

The system can be understood as a coordination platform connecting shelter needs, warehouse resources, volunteers, and relief deliveries. The Disaster Manager is the main operational actor, while the Relief Volunteer interacts with assigned work. The requirements and UML model should remain traceable to these responsibilities and to the problem statement's three central concerns: supplies, volunteer skills, and vehicle deliveries.